

Approximate Entropy (ApEn).

- a statistical measure used to quantify the **regularity and unpredictability** of a time series.

```
def ApEn(U, m, r):  
    """Compute Aproximate entropy"""  
    def _maxdist(x_i, x_j):  
        return max([abs(ua - va) for ua, va in zip(x_i, x_j)])  
    def _phi(m):  
        x = [[U[j] for j in range(i, i + m - 1 + 1)] for i in range(N - m + 1)]  
        C = [len([1 for x_j in x if _maxdist(x_i, x_j) <= r]) / (N - m + 1.0) for x_i in x]  
        return (N - m + 1.0)**(-1) * sum(np.log(C))  
    N = len(U)  
    return abs(_phi(m+1) - _phi(m))
```

- **Low Approximate Entropy** → data is regular and predictable.
- **High Approximate Entropy** → data is irregular and unpredictable.
- **Here, U:** Input time series;
 m: Length of the pattern to compare
 r : Tolerance

- Computes the **maximum absolute distance** between two sequences using

def _maxdist(x_i, x_j):

- Phi computes the quantity, which measures how frequently patterns of length m repeat by creating a sequence and comparing it with every other sequences
- Compute the logarithm using

sum(np.log(C))

- Because, Entropy is based on logarithms and the log converts probabilities into an information measure.
- Now average the function. So

return (N - m + 1.0)(-1) * sum(np.log(C))**

becomes like,

$$\text{phi}(m) = \text{average}(\log(C))$$

- **N=len(U)** Stores the total number of observations.
- **return abs(_phi(m+1)-_phi(m))** compares the **m** value and **m+1** value to find, if two sequences are similar for length **m** or not
- If yes → predictable, low entropy
- If no → unpredictable, high entropy

Approximate Entropy	Meaning
Close to 0	Very regular, highly predictable
0.2–0.5	Moderately regular
> 1	Highly irregular, less predictable

Sample Entropy (SampEn)

- This function computes Sample Entropy (SampEn), which measures the complexity and irregularity of a time series
- Like Approximate Entropy (ApEn), it tells you how predictable a dataset is, but **Sample Entropy is an improved version** because it avoids counting self-matches, making it less biased.
- **Low Sample Entropy** → data is regular and predictable.
- **High Sample Entropy** → data is irregular and unpredictable.
- All the steps are same like ApEn, but Instead of probabilities, Sample Entropy simply counts all matching pairs.

Sample Entropy	Meaning
Close to 0	Very regular, highly predictable
0.2–0.5	Moderate, Some irregularity
> 1	Highly irregular and complex